Renal Anatomy, Nephrology & Transplantation Rapid Revision

Campbell-Walsh-Wein · Chapters 88, 90–92 (with 61)

For MCh Urology theory & viva · NEET-SS / competitive MCQs

How to use (total ≈ 2h 20min even at a slow, careful pace)

- Each chapter reads top-to-bottom in the order you would write an essay answer: definition → pathology → diagnosis → management → complications → follow-up. The same material serves theory and viva.
- Boxed **ALGORITHM** and shaded **KEY POINT** panels carry the reasoning examiners ask about ("why do we…?").
- **MCQ one-liners** end each chapter: re-read these the day before the paper.
- The last section puts the big treatment choices side by side for essay questions.

Conventions: **underlined bold** = number to memorise · MCQ = exam trivia only · → *Ch 91* = cross-reference · † = not in the source chapter notes (added because it is essential in practice). Abbreviations are spelled out at first use in each chapter.

Chapter	Topic	Slow-reader time
88	Surgical, Radiologic and Endoscopic Anatomy of the Kidney and Ureter	23 min
90	Fundamentals of Nephrology	52 min
91	Renal Transplantation	25 min
92 / 61	Urologic Complications of Renal Transplantation; Paediatric Transplant	33 min
★	Putting it together: kidney function and transplant decisions	7 min
	Total (9842 words, at 70 words/min)	**2h 20min**

88 Surgical, Radiologic and Endoscopic Anatomy of the Kidney and Ureter

Reading time ≈ 23 min (slow pace)

1 · KIDNEY: POSITION AND RELATIONS

- Retroperitoneal; length ≈**11 cm**, weight 125–170 g (male). Lies T12–L3; **right 1–2 cm lower** (liver); descends ≈3 cm on inspiration and standing.
- **Orientation follows the cone-shaped psoas**: upper poles medial and posterior; hilum rotated anteriorly, kidney angled 30–50° behind the coronal plane. Hence the lower pole lies lateral and anterior for percutaneous access.

Aspect	Right kidney	Left kidney
Superior	Liver; hepatorenal ligament tears liver capsule on traction	Spleen; splenorenal ligament tears splenic capsule
Medial	Duodenum (2nd part), pancreatic head	Pancreatic tail, splenic vessels
Anterior	Hepatic flexure	Splenic flexure, jejunum
Posterior	Psoas, quadratus lumborum; subcostal, iliohypogastric, ilioinguinal nerves; diaphragm, pleura	

Percutaneous hazards

- **Pleura** reaches the 12th rib (lung only to the 10th intercostal space): access above the 11th rib risks pleural and lung injury — prefer subcostal or 11th intercostal space.
- **Liver/spleen**: injury risk rises with puncture in inspiration and organomegaly.
- **Retrorenal colon**: commoner at the lower pole and far commoner prone than supine.
- Upper-pole tumour excision through the diaphragm → pleural tear, pneumothorax.

2 · COVERINGS AND COMPARTMENTS

- Inside out: fibrous capsule → perirenal fat (enters the sinus) → renal (Gerota) fascia enclosing kidney and adrenal → paranephric fat.
- **Gerota fascia is a cone closed above and laterally, open below**: it contains tumour and perinephric collections, yet fluid can track down into the pelvis. Each perirenal space is separate, so extravasation after endourology rarely reaches the other side; the anterior pararenal space (colon, duodenum, pancreas) crosses the midline. Posterior pararenal space holds fat only.
- Transperitoneal access: incise the **white line of Toldt**; right hilum — **Kocher manoeuvre**; left hilum — mobilise pancreatic tail, spleen and splenic vessels medially.

3 · PARENCHYMA AND ITS IMAGING

- Cortex ≈1 cm; 8–18 pyramids, each papilla cupped by a minor calyx. **Columns of Bertin** carry interlobar arteries, so the safe track is **through pyramid and papilla**.
- Glomerular filtration rate (GFR) ≈**125 mL/min**; ≈99% of filtrate reabsorbed. Urothelium lines calyces to urethra → urothelial carcinoma at any level.
- Hilum anterior → posterior: **vein, artery, pelvis, posterior segmental artery**.

Modality	Normal kidney
Ultrasonography (US)	Cortex iso-/hypoechoic to liver (iso-/hyperechoic in newborn); pyramids hypoechoic; sinus hyperechoic
Computed tomography (CT)	Unenhanced 30–60 Hounsfield units (HU), cortex = medulla; enhanced 80–120 HU
Magnetic resonance imaging (MRI)	T1 cortex brighter; T2 cortex slightly darker

CT phase	Timing	Use
Arterial	20–30 s	Vessels
Corticomedullary	30–70 s	Cortex enhances
Nephrographic	**80–120 s**	**Best for renal masses**
Excretory	>3 min	Collecting system, ureter

- Intravenous urography (IVU): nephrotomogram 60–90 s, panoramic 15 min; no excretion at 24 h = non-functioning kidney.
- **Corticomedullary differentiation** fades in chronic kidney disease and acute kidney injury and returns with recovery; parenchymal volume correlates with function better than length.

Congenital anomalies

- **Malrotation**: pelvis central; calyces medial to the pelvis are the hallmark. **Ectopia**: ureter length suits the position but drainage is poor → stasis, infection, stones; aberrant arteries from adjacent vessels.
- **Crossed fused ectopia**: the crossed kidney's ureter crosses back to a normal orifice.
- **Horseshoe**: lower poles fused; ascent arrested by the **inferior mesenteric artery (IMA)**; associated ureteropelvic junction obstruction (UPJO), stones, infection, multiple arteries.

4 · RENAL VESSELS, LYMPHATICS AND NERVES

- Arteries arise at L1–L2; right passes behind the inferior vena cava (IVC).
- **Five segmental end arteries** (apical, upper, middle, lower from the anterior division; posterior). No anastomosis → occlusion infarcts a segment, but segments can be resected independently.
- **Posterior segmental artery**: first branch, behind the pelvis; if anterior to the ureter → UPJO.
- Sequence: segmental → lobar → interlobar → arcuate (corticomedullary junction) → interlobular → afferent arteriole.
- **Accessory arteries 25–28%** are the sole supply of their territory (usually lower pole): may contraindicate laparoscopic donor nephrectomy and bleed at endopyelotomy → *Ch 91*.
- **Brödel line**: avascular plane just behind the lateral border; site varies, so map by angiography or segmental methylene blue. Used for anatrophic nephrolithotomy (Boyce).
- **Veins anastomose freely** (infundibular collars, subcapsular plexus) → segmental venous occlusion is harmless.

Feature	Right renal vein	Left renal vein
Length	2–4 cm	**6–10 cm**
Tributaries	None	Adrenal, gonadal, lumbar (easily avulsed)
Course	Direct to IVC	Between superior mesenteric artery (SMA) and aorta → **nutcracker syndrome**
Anomaly	Accessory veins	**Circumaortic** (commonest); retroaortic

- **CT angiography** is the gold standard for vascular anatomy (misses vessels <2 mm).
- Lymphatics: left kidney → left para-aortic; right kidney → interaortocaval and paracaval, **may cross to left para-aortic**.
- Sympathetic T8–L1; the kidney functions without nerves (transplants prove it).

5 · PERCUTANEOUS PUNCTURE AND VESSELS

Route (Sampaio endocasts)	Vascular injury
Upper-pole infundibulum	**67%** — most dangerous
Mid-pole infundibulum	Arterial 23%
Lower-pole infundibulum	Artery present in 38%

Route (Sampaio endocasts)	Vascular injury
Through-and-through	Anterior segmental vessels → marked haemorrhage
Calyceal fornix	Venous <8%, **no arterial**

KEY POINT Why the fornix? Arteries run on the anterior and posterior faces of the infundibula; a forniceal puncture through the papilla avoids them. If an infundibulum is unavoidable, choose the lower pole. Consequences of vascular injury: haemorrhage, arteriovenous fistula, pseudoaneurysm, loss of renal function.

6 · PELVICALYCEAL SYSTEM

- 7–9 calyces; compound calyces at the poles; pelvis 3–10 mL.
- **Duplex pelvis**: upper and lower parts do not connect — suspect when a pole cannot be reached from the puncture; confirm on excretory CT urography or retrograde pyelography.
- CT urography has replaced IVU; magnetic resonance urography static-fluid phase suits the dilated system.
- Dilatation mimics: extrarenal pelvis; **megacalycosis** (non-obstructive, extra calyces, normal pelvis); calyceal diverticulum (from fornix into cortex).

Sampaio type	%	Midzone drains
A-I	45	Via a polar group
A-II	17.2	**Crossed calyces** to both polar groups
B-I	21.4	Independent major calyx
B-II	16.4	Minor calyces direct to pelvis

- **Perpendicular calyx** (11.4%): hidden on pyelogram; narrow infundibulum hampers lithotripsy clearance and percutaneous access.
- **Crossed calyces**: limb to the inferior group ventral in **87.5%** — relevant when entering the pelvis through it at percutaneous nephrolithotomy or at retrograde ureterorenoscopy; an interpelviocalyceal region on fluoroscopy is the indirect sign.

7 · URETER: COURSE AND RELATIONS

- Adult 22–30 cm (neonate 6.5–7 cm); lies just lateral to the lumbar transverse process tips. **Narrowings** (stone impaction): ureteropelvic junction (UPJ), iliac crossing, ureterovesical junction (UVJ).
- Descends on psoas, crossing the **common iliac bifurcation** — landmark to find it.
- Gonadal vessels cross anteriorly. Right: ileum, appendix nearby; left: sigmoid mesocolon. **Pelvic brim** is a commonest injury site.
- **Pelvic ureter** (≈15 cm): towards the ischial spine along the greater sciatic notch → turns medially → crosses obturator vessels and nerve → bladder.
- Female: posterior boundary of the ovarian fossa → under the **uterine artery 1.5–2 cm** from cervix ("water under the bridge"); left more at risk in vaginal surgery. Male: vas loops over it.
- **Intramural** 1.2–2.5 cm within Waldeyer sheath; bladder pressure closes the oblique tunnel → antireflux.

8 · RADIOLOGIC ANATOMY OF THE URETER

- **Peristalsis**: whole ureter rarely opacified; a continuous column suggests obstruction or atony.
- Stricture requires **proximal dilatation**; crossing vessels and atherosclerotic iliac indentation mimic it.
- Prostatic enlargement → "fish-hook" distal ureter.

- Duplication: incomplete (one bud splits; one orifice) commoner than complete (two buds, two orifices).

> **KEY POINT** **Weigert–Meyer rule**: upper-moiety orifice **inferior and medial** → ureterocele, obstruction; lower moiety → vesicoureteral reflux (VUR).

- Medial displacement — bilateral: retroperitoneal fibrosis, pelvic lipomatosis, after abdominoperineal surgery; unilateral: enlarged hypogastric nodes, bladder diverticulum, hypogastric artery aneurysm.
- Image in expiration (inspiration kinks the ureter) and after voiding (to see the distal ureter).

9 · URETERIC BLOOD SUPPLY, NERVES, HISTOLOGY

Segment	Artery	Enters	Endoureterotomy
Proximal	Renal	Medially	**Lateral**
Middle	Common iliac	Posteriorly	**Anterior**
Distal	Superior vesical	Laterally	**Medial**

- Veins drain to renal, gonadal and iliac veins. Lymph: abdominal ureter → para-aortic/paracaval; pelvic ureter → iliac nodes.
- The longitudinal anastomosis is fragile: preserve adventitia, avoid lateral retraction → prevents ischaemic stricture. Control the obliterated umbilical artery before mobilising the terminal ureter.
- **Myogenic peristalsis** from minor-calyx pacemakers: excised ureter keeps contracting and denervation does not cause reflux. Nerves carry pain with the sympathetics (T11–L1; S2–S4 lower ureter), referred to flank, groin, scrotum or labia.
- **UPJO** histology: excess longitudinal muscle and collagen → functional obstruction.

10 · ENDOSCOPIC ANATOMY

- Trigone: apex at bladder neck, base the interureteric ridge (Mercier bar); most vascular part of the bladder. Orifices 1–2 cm from midline, pushed laterally as the bladder fills.

Lyon axis	Grades
Position	A medial trigone (normal) · B · C lateral
Configuration	0 volcano (normal) · 1 stadium · 2 horseshoe · 3 golf-hole

- Higher grade → more reflux.
- **Intramural ureter is narrowest** (3–4 mm); may need dilatation.
- Brim and UPJ look stenotic but open with irrigation; iliac pulsation marks the brim, where the ureter angles posteriorly.
- Advance over a safety guidewire following that course; wait for the peristaltic wave to relax; reduce tidal volume to limit renal excursion.
- Lower-pole infundibulum needs ≈**140°** deflection. Papillae: pale discs ringed by fornices; ducts of Bellini dilate with obstruction.

ALGORITHM
Infundibulotomy: superior → inferior → posterior quadrants; **never anterior**

11 · MCQ ONE-LINERS

1. Hilum: vein, artery, pelvis, posterior segmental artery.
2. Gerota fascia open inferiorly.
3. Supracostal access above 11th rib risks pneumothorax.
4. Best CT phase for renal mass: nephrographic.
5. Horseshoe ascent arrested by IMA.
6. Five segmental end arteries.
7. Accessory renal arteries: 25–28%.
8. Commonest left renal vein anomaly: circumaortic.
9. Safest puncture: calyceal fornix.

10. Sampaio A-II: crossed calyces.
11. Weigert–Meyer: upper moiety inferomedial.
12. Distal ureteric endoureterotomy: medial.
13. Peristalsis is myogenic.
14. Lyon golf-hole: grade 3.
15. Infundibulotomy: never anterior.
16. Narrowest ureter: intramural.
17. Right kidney 1–2 cm lower than left.
18. Anterior pararenal space crosses the midline.
19. Interlobar arteries run in columns of Bertin.
20. Posterior segmental artery anterior to ureter causes UPJO.
21. Left renal vein: 6–10 cm, three tributaries.
22. Nutcracker: left renal vein between SMA and aorta.
23. Most dangerous puncture: upper-pole infundibulum.
24. Ureteric narrowings: UPJ, iliac crossing, UVJ.

90 Fundamentals of Nephrology

Reading time ≈ 52 min (slow pace)

1 · WHY A UROLOGIST NEEDS NEPHROLOGY

- **Nephron**: ≈**1 million** per kidney; glomerulus + tubule (proximal tubule → loop of Henle → distal tubule → collecting duct). Number fixed at birth; reduced nephron mass (nephrectomy, agenesis) → **hyperfiltration** of the remainder.
- **Filtration barrier** = endothelium + glomerular basement membrane (GBM) + podocytes; selects by size and charge.
- Functions: waste/drug excretion; sodium (= volume) and blood pressure via the renin–angiotensin–aldosterone system (RAAS); water, electrolyte and acid–base balance; **erythropoietin** (interstitial fibroblasts, hypoxia-driven); **calcitriol** (proximal tubule 1α-hydroxylation, stimulated by parathyroid hormone [PTH], low calcium or phosphate).

> **KEY POINT** Whoever operates on, obstructs, relieves or doses drugs through the kidney is changing volume, electrolytes and acid–base balance. The glomerulus is a pressure-driven filter set by arteriolar tone; the tubule is an energy-hungry line working at the edge of hypoxia — most disorders below fail one or the other.

2 · MEASURING KIDNEY FUNCTION

Steady state — glomerular filtration rate (GFR)

- GFR cannot be measured directly. Gold standard: clearance of inulin, iohexol or iothalamate — cumbersome.
- **Serum creatinine (SCr)**: muscle product, freely filtered, plus tubular secretion that grows as function falls. SCr alone misleads because generation depends on muscle mass.
- **24-hour creatinine clearance overestimates** GFR (secretion) and suffers collection errors.
- **Estimated GFR (eGFR)**: laboratories should report the **2021 Chronic Kidney Disease Epidemiology Collaboration (CKD-EPI) creatinine equation, without race** (age and sex only). The old race coefficient raised eGFR in Black patients, delaying referral.
- **Cystatin C**: 13-kDa protein, freely filtered, completely reabsorbed and degraded proximally; never race-dependent. The **combined creatinine–cystatin C equation is most accurate**; use when creatinine eGFR is likely biased or a decision sits at a staging threshold.

Error source	Serum creatinine	Cystatin C
Generation ↑	Large muscle mass, high-protein diet, cooked meat, creatine supplements	Hyperthyroidism, glucocorticoids, obesity, inflammation, smoking
Generation ↓	Small muscle mass, amputation, muscle wasting	Hypothyroidism
Secretion blocked (level ↑)	**Trimethoprim, cimetidine**, fenofibrate	—
Extrarenal (level ↑)	**Urine leak with peritoneal reabsorption**	—
Extrarenal loss (level ↓)	Dialysis; large pleural/ascitic drainage	Large pleural/ascitic drainage

> **KEY POINT** Trap: after cystectomy or bladder repair, rising SCr with ascites may be **intraperitoneal urine leak** reabsorbed across the peritoneum — not kidney injury.

Urinary protein

- Normal excretion <**150 mg/day** (barrier limits passage; proximal tubule reclaims). More = barrier/tubular damage or overproduction (myeloma paraproteins). †Nephrotic range >3.5 g/day.

- Albuminuria is the sensitive marker of glomerular injury and independently predicts CKD progression, cardiovascular disease, stroke and death.
- Kidney Disease: Improving Global Outcomes (KDIGO) 2023 hierarchy: **spot albumin-to-creatinine ratio (ACR)**, early morning → protein-to-creatinine ratio (PCR) if ACR unavailable or non-albumin protein (Bence Jones) suspected → reagent strip, confirmed formally.
- Dipstick detects **albumin only — misses light chains**.

Dynamic state

- **eGFR equations are invalid when function is changing** — an anuric patient has GFR zero whatever the eGFR reads. Acute kidney injury (AKI) is therefore defined by SCr change and urine output.
- Both are late; novel biomarkers remain investigational.

3 · ACUTE KIDNEY INJURY

Glomerular haemodynamics

- Net filtration pressure ≈**10 mm Hg** = glomerular hydrostatic 60 − Bowman's 18 − oncotic 32. GFR depends on holding glomerular hydrostatic pressure by adjusting afferent and efferent tone.

Arteriolar change	Renal blood flow	GFR
Afferent constriction	↓	↓
Afferent dilatation	↑	↑
Efferent constriction	↓	↑
Efferent dilatation	↑	↓

Efferent constriction lowers flow yet raises GFR — so angiotensin II blockade drops GFR when perfusion is low.

Definition and staging (KDIGO)

Stage	Serum creatinine	Urine output
1	Rise ≥**0.3 mg/dL** (within 48 h†) or ×1.5 baseline (within 7 days†)	<0.5 mL/kg/h for 6 h
2	×2 baseline	<0.5 mL/kg/h for 12 h
3	×3, or SCr ≥4 mg/dL with acute rise ≥0.5 (†or dialysis started)	<0.3 mL/kg/h for 24 h, or anuria 12 h

- Either criterion suffices. **3 months** divides acute kidney disease (AKD) from chronic kidney disease (CKD).
- AKI affects one-fifth to one-third of inpatients; mortality ≈1 in 4, ≈1 in 2 if renal replacement therapy (RRT) needed. Prerenal AKI + acute tubular necrosis (ATN) ≈ three-quarters; often multifactorial.

Category	Compartment	Causes
Prerenal	Systemic perfusion	Hypovolaemia, cardiac failure, hepatorenal syndrome
Intrinsic	Vessels	Renal artery/vein occlusion, vasculitis, atheroembolism
Intrinsic	Glomerulus	Anti-GBM disease, lupus, postinfectious, immunoglobulin A (IgA) nephropathy
Intrinsic	Tubule (ATN)	Ischaemia, nephrotoxins, rhabdomyolysis, contrast
Intrinsic	Interstitium (acute interstitial nephritis, AIN)	Drugs, infection, systemic disease
Intrinsic	Intratubular	Myeloma cast nephropathy, drugs, crystals

Category	Compartment	Causes
Postrenal	Outflow	Bladder outlet obstruction, tumours, calculi, papillary necrosis, retroperitoneal fibrosis

Prerenal AKI

- Volume depletion → reduced perfusion → fall in glomerular hydrostatic pressure. Structure normal: **bland urine, low fractional excretion of sodium (FeNa), immediate recovery** on correction.
- **Autoregulation** keeps GFR constant over a wide mean arterial pressure (MAP) range by afferent dilatation + efferent constriction:
  - **Myogenic reflex**: afferent smooth muscle senses pressure; dilatation is prostaglandin-mediated → **disabled by non-steroidal anti-inflammatory drugs (NSAIDs)**.
  - **Tubuloglomerular feedback (TGF)**: macula densa senses low NaCl → renin → angiotensin II → efferent constriction → **disabled by RAAS blockade**.
- Autoregulation also impaired by age, CKD, transplant, vascular disease, sepsis, hypercalcaemia and contrast.
- **Modifiable risk factors** (the surgeon's levers): hypotension, anaemia, sepsis, nephrotoxins, rhabdomyolysis, hypercalcaemia.

Index	Prerenal	ATN
Urine sodium	<20 mmol/L	>20 mmol/L
FeNa	**<1%**	>1%
Fractional excretion of urea (FeUrea) — on diuretics/alkalosis	**<35%**	>35%
Specific gravity	>1.015	<1.010
Sediment	Bland (hyaline casts allowed)	Tubular cells, **muddy-brown granular casts**
After removing insult	Immediate recovery	SCr keeps rising, plateaus, recovers over days–weeks

- Prerenal indices show working tubules defending volume. High FeNa after new diuretics does not prove ATN — use FeUrea. **Response to correction is the single most important test.**
- **Management**: hypovolaemia → isotonic fluid; heart failure → diuretics; postoperative hypotension → hold antihypertensives. MAP **>65 mm Hg** suffices in healthy adults, but elderly vasculopaths autoregulate at higher pressures.

Fluids (effective osmolarity, mOsm/L): plasma 308; 0.9% saline 308 (Na 154); Plasmalyte 294; lactated Ringer's 273 (Na 130, K 4); 0.45% saline 154; 5% dextrose (D5) 0 — dextrose is metabolised, so D5 is free water and a D5–saline behaves as its saline base.

Intrinsic AKI — the cast localises the compartment

	ATN	Acute glomerulonephritis	AIN
Cause	Ischaemia, toxins	Immune complex, anti-GBM, cell-mediated	Drug, infection, systemic
Urine	Granular (muddy-brown) casts	Dysmorphic red cells, **red cell casts**, proteinuria	White cells, **white cell casts**, ± eosinophils

Acute tubular necrosis

- **Why tubules**: despite ≈25% of cardiac output, countercurrent vasa recta flow drops medullary oxygen tension to 10–20 mm Hg, while the **S3 proximal segment and medullary thick ascending limb** consume heavily. Hospital ATN is usually mixed ischaemic + toxic; cell death is mostly apoptosis.

- **Mechanism**: ischaemia displaces Na^+/K^+-ATPase from the basolateral membrane → proximal sodium reabsorption fails → high macula densa NaCl → TGF afferent constriction → GFR falls. This fall is **protective**: it cuts tubular workload while cells regenerate. Sloughed cells + Tamm–Horsfall protein = muddy-brown casts.
- **Rhabdomyolysis** (obese patient, long flexed lateral decubitus robotic case; may be painless): myoglobin → toxicity, intratubular precipitation, hypoperfusion. Treat with **early aggressive isotonic fluid**; alkalinisation unproven. †Prevent by padding, limiting flexion and operative time.
- **Management**: remove insults — avoid hypotension and nephrotoxins; stop RAAS blockers and NSAIDs unless essential. **Diuretics only for symptomatic volume overload; they do not treat ATN.**

Postcontrast AKI (contrast nephropathy)

- SCr rise within **48 h** of intravenous iodinated contrast (usually computed tomography [CT]): vasoconstriction + direct tubulotoxicity. Risk: age, diabetes, CKD, anaemia, volume, arterial route.
- **Prevention: intravenous fluid before contrast** in hospitalised at-risk patients — the only effective measure; best fluid and rate unclear. †Isotonic saline or bicarbonate are equivalent and N-acetylcysteine adds nothing (PRESERVE). †Radiology–nephrology consensus: prophylaxis for eGFR <30 or AKI; use the lowest diagnostic dose.
- Post-contrast dialysis is never needed, even in dialysis patients.
- **Gadolinium** (magnetic resonance imaging [MRI]): not nephrotoxic at usual dose but groups I and III linked to **nephrogenic systemic fibrosis (NSF)** — avoid in AKI, eGFR **<30**, dialysis. Group II (gadobenate, gadoteridol, gadobutrol, gadoterate) may be used. **Group III (gadoxetate) is treated as group I**, not intermediate.

Acute interstitial nephritis

- Cell-mediated hypersensitivity. **Drugs cause most** — those urologists prescribe daily: β-lactams, fluoroquinolones, sulfonamides, vancomycin, NSAIDs, proton pump inhibitors, allopurinol, **immune checkpoint inhibitors** (commonest biopsy lesion in AKI on these). Also complicated urinary tract infection.
- Onset days–weeks after the drug. Consider in unexplained hospital AKI and **sterile pyuria**. Urine eosinophils are insensitive and non-specific MCQ.
- Biopsy: infiltrate, tubulitis, eosinophils. **Stop the drug**; early corticosteroids help (observational). Only ≈half recover fully.

Acute glomerulonephritis and postrenal AKI

- Glomerulonephritis is in the differential of microhaematuria; acute SCr rise, proteinuria, red cell casts, hypertension or purpura → **urgent nephrology review and biopsy**, not cystoscopy first (rapidly progressive glomerulonephritis).
- Nephrotic diseases present insidiously with proteinuria. Postrenal AKI — section 8.

Complications of AKI

- **Volume overload**: urgent only with respiratory distress. Salt/fluid restriction + high-dose intravenous loop diuretic; escalate, redose by half-life; add metolazone or acetazolamide (**sequential blockade**). Oral furosemide ≈50% bioavailable (oral dose = 2 × intravenous); bumetanide/torsemide equal by either route; torsemide lasts longest (8–12 h).
- **Uraemia**: metallic taste, anorexia, somnolence, asterixis. Uraemic bleeding → **desmopressin (DDAVP) 0.3 µg/kg** intravenously or intramuscularly 30–60 min before procedure; tachyphylaxis after 3 doses; watch hyponatraemia.

Renal replacement therapy — "AEIOU"

Indication (refractory to medical therapy)	Feature
Acidosis	Unresponsive to bicarbonate

Indication (refractory to medical therapy)	Feature
Electrolytes	Mostly hyperkalaemia
Ingestions	Salicylates, alcohols, lithium
Overload	Respiratory distress despite diuretics
Uraemia	Encephalopathy, pericarditis

- AEIOU marks failure of medical management, not a biochemical threshold. Intermittent haemodialysis via 10–12 Fr dual-lumen jugular/femoral catheter; **diffusion** clears solute, **convection** removes fluid. Continuous RRT if haemodynamically unstable. **Early dialysis gives no benefit; continuous is not superior.**
- Non-recovery persisting ≥**90 days** = CKD or end-stage kidney disease (ESKD).

4 · DRUG DOSING IN REDUCED KIDNEY FUNCTION

- Dose on steady-state eGFR; in AKI eGFR overestimates GFR — reassess daily. †Many drug labels were derived with **Cockcroft–Gault creatinine clearance** = (140 − age) × weight (kg) / (72 × SCr mg/dL), × 0.85 if female.
- Avoid NSAIDs; †withhold RAAS blockers during dehydration or perioperative hypotension ("sick-day" rule).
- †Renally cleared drugs needing adjustment in urology: aminoglycosides and vancomycin (level-guided), fluoroquinolones, cephalosporins, low-molecular-weight heparin. †Nitrofurantoin is ineffective and toxic at low GFR — avoid if eGFR <30 (short courses allowed at 30–44 in United Kingdom guidance). †Metformin: stop at eGFR <30; withhold before iodinated contrast in AKI or eGFR <30.

5 · CHRONIC KIDNEY DISEASE

- **Definition (KDIGO)**: abnormal kidney structure or function for ≥**3 months**. Structure alone qualifies — a congenitally atrophic kidney with normal GFR is CKD. Classify by **cause, GFR category, albuminuria category**.

GFR (mL/min/1.73 m²)	A1 ACR <30 mg/g (<3 mg/mmol)	A2 30–300	A3 >300 (>30 mg/mmol)
G1 ≥90	Low*	Moderate	High
G2 60–89	Low*	Moderate	High
G3a 45–59	Moderate	High	Very high
G3b 30–44	High	Very high	Very high
G4 15–29	Very high	Very high	Very high
G5 <15 (kidney failure)	Very high	Very high	Very high

*G1–G2 are CKD only with albuminuria or structural damage. Albuminuria can move a patient two risk bands without any GFR change. Example: eGFR 43, ACR 1100 mg/g = **G3bA3, very high risk**.

- Causes: diabetes, hypertension; also glomerular disease, **stones, chronic obstruction**, genetic. **Each perioperative AKI episode is a modifiable risk factor for progression**, as are albuminuria, smoking and acidosis.

Complications

Complication	Mechanism	Management
Anaemia (normocytic)	Erythropoietin falls, especially GFR <30	Replete iron, exclude other causes → erythropoiesis-stimulating agent; target haemoglobin **10–11.5 g/dL**; >13 harmful

Complication	Mechanism	Management
Secondary hyperparathyroidism (HPT)	Phosphate retention, low calcitriol, low calcium	Phosphate restriction, binders with meals, calcitriol
Tertiary HPT	Autonomous gland, hypercalcaemia	Calcimimetic → subtotal parathyroidectomy
Acidaemia	Low ammoniagenesis, bicarbonate loss	Alkali to keep bicarbonate ≥**22** — slows progression, helps bone
Hyperkalaemia	Low distal sodium delivery; RAAS blockade	Binders + diuretics, preserving RAAS blockers and diet
Hypertension	Sodium/volume retention	Scheduled diuretic; chlorthalidone works in advanced CKD
Erectile dysfunction	Vascular, neuropathy, hypogonadism	Phosphodiesterase type 5 inhibitors; transplant

Slowing progression

- Glycated haemoglobin (HbA1c) <7.5%; blood pressure <**130/80**. Agent matters only in proteinuric disease → RAAS blockade.
- **Sodium–glucose cotransporter-2 (SGLT2) inhibitors**: reno- and cardioprotection with mortality benefit, diabetic and non-diabetic. **Finerenone** (non-steroidal mineralocorticoid antagonist) in CKD with type 2 diabetes.
- Progression is non-linear — drops follow sepsis, nephrotoxins, surgical complications. Calculators predict dialysis after (partial) nephrectomy from baseline function.

6 · HAEMATURIA AND PROTEINURIA — WHEN TO INVOLVE A NEPHROLOGIST

- †Microhaematuria = ≥**3** red cells/high-power field on microscopy; a positive dipstick needs microscopic confirmation.
- **Glomerular clues**: dysmorphic red cells, red cell casts, proteinuria (raised ACR/PCR), rising SCr, hypertension, purpura or systemic disease. These redirect work-up to nephrology.
- †Glomerular disease does not exclude urothelial cancer; urological evaluation still applies when risk factors exist.

ALGORITHM
Haematuria ± proteinuria → urine microscopy + ACR + SCr/eGFR → red cell casts, dysmorphic cells, AKI or systemic features → **urgent nephrology**; isolated non-glomerular haematuria → urological evaluation; isolated albuminuria → stage CKD, refer if criteria below.

- †KDIGO referral criteria: AKI or abrupt sustained fall in GFR; GFR <30; persistent ACR ≥300 mg/g (≥30 mg/mmol); progression (sustained fall >5 mL/min/1.73 m²/year); red cell casts or unexplained persistent haematuria >20 red cells/high-power field; hypertension refractory to ≥4 agents; persistent potassium abnormality; recurrent or extensive nephrolithiasis; hereditary kidney disease.

7 · ELECTROLYTE AND ACID–BASE DISORDERS

Water and sodium

- **Sodium disorders are water disorders**: low sodium = water excess; high = water deficit. Volume status is a separate, clinical diagnosis; urine sodium is an adjunct (low = low effective volume).
- Water reclamation needs a concentrated medulla + **apical aquaporin 2 (AQP2)**, inserted by antidiuretic hormone (ADH) via the V2 receptor. ADH responds to osmolality and, more potently, volume — **volume depletion overrides hypo-osmolality**.

- Total body water = 0.6 × weight; ⅔ intracellular; extracellular ¾ interstitial, ¼ intravascular. Of 1 L: D5 → **83 mL** intravascular; 0.9% saline → 250 mL; 5% albumin → 1000 mL.

Hyponatraemia

- Confirm hypo-osmolality first. Hyperosmolar: hyperglycaemia (correct sodium). †Urological: absorption of hypotonic glycine/sorbitol irrigant during monopolar transurethral resection (transurethral resection syndrome).

Volume	Urine	Causes
Hypovolaemic	Na >20	Diuretics, mineralocorticoid deficiency, osmotic diuresis, cerebral salt wasting
Hypovolaemic	Na <20	Vomiting, diarrhoea
Euvolaemic	Osmolality > serum	Syndrome of inappropriate ADH (SIADH), glucocorticoid deficiency, hypothyroidism, pain, nausea, drugs
Euvolaemic	Osmolality low	Polydipsia, low-solute intake ("tea and toast", beer potomania)
Hypervolaemic	Na >20	Kidney failure
Hypervolaemic	Na <20	Heart failure, cirrhosis, nephrotic syndrome

- SIADH is a diagnosis of exclusion → water restriction, then vaptans, salt + furosemide, urea. Hypovolaemia → isotonic fluid.
- **Rate**: chronic/unknown onset → rise **≤8–12 mEq/L per 24 h**, else **osmotic demyelination**. Highest risk: fluid-resuscitated hypovolaemia removes the ADH stimulus → brisk water diuresis; overcorrection → DDAVP + free water. Severe symptoms (seizure, obtundation) → **3% saline to raise 4–6 mEq/L** over hours, then slow.

Hypernatraemia

- Water deficit. Diabetes insipidus (DI): central or nephrogenic (lithium, hypercalcaemia, hypokalaemia, **post-obstruction**). Thirst + water access prevents it — the nil-by-mouth postoperative DI patient is at risk.
- Free water; chronic → fall **≤0.5 mEq/L/h** (else cerebral oedema). Hypo-correction shrinks brain cells; hyper-correction swells them.

Potassium

- 98% intracellular. Defence: cell shift (insulin, β_2-agonists), then distal secretion — increased by urine flow, distal sodium delivery, aldosterone, alkalosis.
- **Hyperkalaemia**: shift (insulin deficiency, acidosis, cell lysis, β-blockers) or reduced excretion (AKI/CKD, aldosterone deficiency, **obstructive uropathy**, drugs). Urological: **jejunal conduit** reabsorbs potassium. Exclude pseudohyperkalaemia (haemolysis, fist clenching, leucocytosis, thrombocytosis).
- Electrocardiogram (ECG): peaked T → long PR, wide QRS → P loss → sine wave → ventricular fibrillation. >**6.0** generally urgent.

ALGORITHM
Stabilise (any ECG change): calcium gluconate 10% 10 mL intravenously → **shift**: insulin 10 U + 50 mL 50% dextrose; nebulised salbutamol 20 mg → **remove**: haemodialysis if oliguric/ESKD; loop diuretic if non-oliguric; bicarbonate if acidotic; potassium binder → stop culprit drugs.

Binders: **sodium polystyrene sulfonate (SPS)** — only one emergency-approved, yet slow and causes **colonic necrosis** (postoperative, ileus) → not first-line; patiromer (≈7 h); sodium zirconium cyclosilicate (≈1 h).

- **Hypokalaemia**: diarrhoea, diuretics, hyperaldosteronism, renal tubular acidosis (RTA), hypomagnesaemia, refeeding. **Vomiting** causes it via alkalosis → bicarbonaturia → distal sodium delivery → kaliuresis, not gastric loss. ECG: flat T, U wave. Replace with potassium chloride, **oral preferred**, 20–40 mEq repeated.

Calcium

- Reabsorption: proximal tubule > thick ascending limb (TAL) — both coupled to sodium — > distal convoluted tubule (uncoupled). Hence saline/loop diuretic → calciuria; **thiazide → reduced urinary calcium** (used in calcium stone formers).
- Corrected calcium: +0.8 mg/dL per 1 g/dL albumin below 4.
- **Hypercalcaemia work-up in a stone former**: PTH with simultaneous calcium → non-suppressed = primary HPT (low phosphate a clue) → suppressed: PTH-related peptide (PTHrP; renal, bladder, squamous cancers) and 1,25-vitamin D (sarcoid, lymphoma) → neither: bone resorption, thiazide. Low urine calcium → familial hypocalciuric hypercalcaemia (benign).
- Treat if symptomatic or ≥**14 mg/dL**: **isotonic saline** 200 mL/h, titrated to urine 100–150 mL/h; loop diuretic only if overloaded; calcitonin (rapid, tachyphylaxis 48 h); intravenous bisphosphonate (sustained); denosumab.

Acid–base

- Metabolic acidosis = bicarbonate <22. Anion gap (AG) = Na − (Cl + HCO_3), raised >12; add 2.5 per 1 g/dL albumin below 4. Compensation never normalises pH.
- In AG acidosis the gap rise should match the bicarbonate fall; a larger fall = coexisting non-AG acidosis.
- **Urine AG** = (Na + K) − Cl: negative = ammonium excreted = gut loss; positive = renal (RTA).

	Proximal (type II)	**Distal (type 1)**	**Type IV**
Defect	Bicarbonate reclamation	Distal H^+ secretion	Hypoaldosteronism → hyperkalaemia → low ammoniagenesis
Potassium	Low (worse with bicarbonate)	Low	High
Urine pH	High on carbonic anhydrase inhibitors	High (>6)	—
Stone link	**Topiramate, zonisamide** → calcium phosphate stones	Hypocitraturia ± hypercalciuria → calcium stones	—

- Distal RTA with hyperkalaemia: **obstructive uropathy**, lupus, sickle cell; drugs blocking the epithelial sodium channel (amiloride, triamterene, trimethoprim).
- Alkali: potassium citrate when hypokalaemia coexists; 1 mmol citrate = 1 mmol bicarbonate.
- MCQ Laxative abuse → negative urine AG, urine pH >5.8 → **ammonium acid urate stones**.
- †Ileal/colonic conduits and neobladders → hyperchloraemic non-AG acidosis (bowel absorbs ammonium chloride).

8 · URINARY TRACT OBSTRUCTION AND POST-OBSTRUCTIVE DIURESIS

Term	**Meaning**
Hydronephrosis	Pelvicalyceal dilatation, with or without obstruction
Obstructive uropathy (OU)	Functional or anatomical obstruction at any level; GFR change depends on the other kidney
Obstructive nephropathy	Irreversible parenchymal loss

- OU is **one of the few curable kidney diseases** — exclude it in any unexplained AKI or CKD (†renal ultrasound). Unilateral OU may be invisible in SCr yet destroys that kidney; bilateral leads to ESKD → *Paediatric Urology book, Ch 51*.

Post-obstructive diuresis

- Physiological after relief of prolonged **bilateral** obstruction (typically catheterising chronic retention). Up to **300 mL/h** common; starts within hours, lasts ≈2 days; usually self-limited.

- **Three mechanisms**: (1) transporter dysregulation — Na^+/K^+-ATPase and aquaporins (chiefly AQP2) suppressed, occasionally for weeks → nephrogenic DI (urine 300–400 mOsm/kg, risk of hypernatraemia); (2) volume expansion released — less sympathetic tone and aldosterone, more atrial natriuretic peptide (ANP); (3) **urea osmotic diuresis**.
- **Gradual bladder decompression has no role** — it reduces neither haematuria nor diuresis.

ALGORITHM
Relieve obstruction → free oral fluids, no routine IV → output persistently high, hypovolaemia or obtunded/nil by mouth → metabolic panel every **4–6 h** (Na, K, phosphate, calcium, magnesium) → replace ≈**75%** of urine output with **0.45% saline** + electrolytes.

KEY POINT Why not replace all output? The diuresis is mostly appropriate excretion of retained salt, water and urea; †full replacement keeps the patient expanded and sustains it.

Recovery after obstruction

Canine study	Model	Recovery
Vaughan and Gillenwater	Complete unilateral ureteral obstruction (UUO)	7 days → 100%; 14 days → 70%; 4 weeks → 30%; **6 weeks → none**

- Injury is chiefly **vascular** (arteriolar vasoconstriction, capillary collapse) before tubular; duration, not degree of dilatation, governs damage.
- Small ureteric stone with low-grade obstruction: observation for **2–4 weeks** is functionally safe. Beyond 4 weeks outcome is unpredictable (recovery reported up to 5 months).
- Favour recovery: lesser obstruction, compliant system, pyelolymphatic backflow, youth, thick cortex — not judgeable acutely → **low threshold for stent or nephrostomy**. †Infection with obstruction demands urgent drainage.

9 · LANDMARK TRIALS

All randomised controlled trials (RCTs).

Trial	Design	Result	Take-home
†PRESERVE	RCT, angiography; bicarbonate vs saline, N-acetylcysteine vs placebo	No difference	Saline; no N-acetylcysteine
†STARRT-AKI	RCT, accelerated vs standard RRT start	No survival benefit; more dialysis dependence	Dialyse for AEIOU, not early
†DAPA-CKD / EMPA-KIDNEY	RCTs, SGLT2 inhibitor vs placebo, CKD ± diabetes	Less progression	Standard in CKD
†FIDELIO-DKD	RCT, finerenone vs placebo, CKD + type 2 diabetes	Fewer kidney and cardiovascular events	Add to RAAS blockade
†CHOIR / TREAT	RCTs, high vs lower haemoglobin target	Harm with high target	Avoid haemoglobin >13 g/dL

10 · MCQ ONE-LINERS

1. Most accurate estimate: combined creatinine–cystatin C equation.
2. Trimethoprim and cimetidine raise SCr by blocking secretion, not GFR.
3. Rising SCr with ascites after bladder surgery: suspect intraperitoneal urine leak.
4. Normal protein excretion <150 mg/day; first-line test: spot ACR.
5. KDIGO AKI stage 1: SCr rise ≥0.3 mg/dL or urine <0.5 mL/kg/h for 6 h.
6. Efferent constriction lowers flow but raises GFR.
7. NSAIDs block afferent dilatation; RAAS blockers block efferent constriction.
8. Prerenal: FeNa <1%; FeUrea <35% on diuretics.
9. Muddy-brown granular casts: ATN; red cell casts: glomerulonephritis; white cell casts: AIN.
10. Diuretics treat overload in ATN, not ATN itself.
11. Contrast AKI prevention: intravenous fluid before contrast in at-risk patients.

2. Gadolinium group III handled as group I; group II allowed in advanced CKD.
3. Commonest biopsy lesion in checkpoint-inhibitor AKI: AIN.
4. Uraemic bleeding: DDAVP 0.3 µg/kg; tachyphylaxis after 3 doses.
5. CKD anaemia target haemoglobin 10–11.5 g/dL.
6. Chronic hyponatraemia: correct ≤8–12 mEq/L/24 h — osmotic demyelination.
7. Chronic hypernatraemia: lower ≤0.5 mEq/L/h — cerebral oedema.
18. Sodium polystyrene sulfonate: only emergency-approved binder; risks colonic necrosis.
19. Vomiting causes hypokalaemia through kaliuresis, not gastric loss.
20. Topiramate: proximal-type RTA, high urine pH, calcium phosphate stones.
21. Negative urine anion gap: gastrointestinal bicarbonate loss.
22. Post-obstructive diuresis: replace ≈75% of output with 0.45% saline if unable to drink.
23. Complete UUO: full recovery at 1 week, none after 6 weeks (canine).

91 Renal Transplantation

Reading time ≈ 25 min (slow pace)

1 · END-STAGE DISEASE AND WHY TRANSPLANT

- Chronic kidney disease (CKD) is defined by reduced estimated glomerular filtration rate (eGFR), albuminuria, or both. Kidney Disease: Improving Global Outcomes (KDIGO) stage **G5 (eGFR <15 mL/min/1.73 m²)** = end-stage renal disease (ESRD), needing dialysis or transplant. CKD staging, dialysis modalities and their complications → *Ch 90*.
- Years on dialysis accrue cardiovascular and mineral–bone damage that the eventual graft inherits; hence **transplantation is the optimal therapy** and **pre-emptive transplantation** (before dialysis) the logical extreme — far commoner with a living donor and in children.
- Transplant may triple life expectancy versus staying on dialysis; better quality of life; cheaper after year one. Permanent arteriovenous (AV) access is still created well ahead of need.
- Risk of death is **higher for the first 3–6 months**, then much lower (see landmark study).
- MCQ First successful kidney transplant: identical twins, 1954.

2 · RECIPIENT EVALUATION

> **KEY POINT** Two questions drive every test: will the patient survive the operation, and will immunosuppression unmask something dormant (infection, cancer)?

- Evaluate before dialysis; deceased-donor listing allowed at eGFR ≤**20**. Elderly may gain from a faster older-donor kidney.
- Commonest adult causes: hypertension and diabetes — control them and lose weight before transplant.

Recurrence risk	Diseases	Implication
Very low	Autosomal dominant polycystic kidney disease (ADPKD), dysplasia, Alport, **genetic** focal segmental glomerulosclerosis (FSGS)	None
Higher	**Primary FSGS**, membranoproliferative glomerulonephritis, membranous nephropathy	Counsel; monitor
Histological often, clinical rarely	Immunoglobulin A (IgA) nephropathy	—
Metabolic	Primary hyperoxaluria	Combined liver–kidney

Logic: disease encoded in native-kidney genes cannot recur in a graft carrying donor genes; disease from circulating or immune factors can.

Medical work-up

- **Cardiac**: coronary artery disease (CAD) is the main cause of death. High-risk asymptomatic (diabetes, smoking, prior CAD, long dialysis) → echocardiography, stress testing, selective angiography; retest long waiters.
- **Vascular**: heavy iliac calcification may preclude anastomosis → Doppler and non-contrast computed tomography (CT) in diabetics, smokers, elderly. Non-healing foot ulcer/osteomyelitis → defer. ADPKD with family history of intracranial bleed → screen for cerebral aneurysm.
- **Obesity** (relative contraindication): more wound complications, delayed graft function (DGF) and rejection. Frailty predicts mortality.
- **Psychosocial**: non-adherence, substance abuse or unstable psychiatric illness must be controlled first.
- **Infection**: screen tuberculosis (TB; active TB contraindicates until treated), hepatitis B and C, human immunodeficiency virus (HIV). Cytomegalovirus (CMV), herpes simplex and Epstein–Barr

virus (EBV) serology directs prophylaxis. **Vaccinate before** — live vaccines are barred afterwards. Direct-acting antivirals allow hepatitis C-viraemic donor kidneys into negative recipients.
- Urologist's share: medication and problem lists, prostate-specific antigen, urinalysis/culture, ultrasound, exclusion of active infection or malignancy and of conditions unsuitable for technical success.

Malignancy	Transplant
Prostate cancer Gleason ≤6; superficial non-melanoma skin cancer; incidental renal tumour <1 cm	Possible without prior treatment
Basal/squamous skin cancer, melanoma in situ, renal cell carcinoma <3 cm, small low-grade thyroid cancer, superficial bladder cancer	No waiting period after cure
All other	After treatment, individualised wait

Surgical considerations

- Long-term femoral dialysis access → exclude iliac venous obstruction; a functioning ipsilateral leg fistula causes graft venous hypertension.
- **Native nephrectomy** before or with transplant: ADPKD needing space or with infection, bleeding, pain; chronically infected kidneys; recurrent urinary tract infection (UTI) with reflux; heavy proteinuria.

3 · IMMUNOLOGICAL COMPATIBILITY

- **ABO**: O universal donor, AB universal recipient. A2 donors may give to B/O recipients with low anti-A titre. ABO-incompatible transplant needs intensified immunosuppression.
- **Human leukocyte antigen (HLA)** compatibility = negative crossmatch (donor lymphocytes + recipient serum). Class I (A, B, C) on all nucleated cells; class II (DR, DQ, DP) on antigen-presenting and endothelial cells.
- **Sensitisation** by transfusion, pregnancy, previous transplant → preformed antibodies → hyperacute or severe rejection. Single-antigen bead (Luminex) assays allow a **virtual crossmatch**.

4 · DECEASED DONATION

- Donation after brain death (DBD) or after circulatory death (DCD). Allocation weighs waiting time (from dialysis start), HLA match, sensitisation, and longevity matching by Kidney Donor Profile Index (KDPI) and Estimated Post-Transplant Survival (EPTS).
- Donor contraindications: active metastatic cancer, haematological malignancy, melanoma history, untreatable viral infection. Treated bacterial infection (even meningitis) is acceptable.
- Paediatric donors **<15 kg → en bloc** kidneys on aortic and caval conduits into an adult with body mass index (BMI) <30 and low immunological risk; thrombosis 10–15%, but long-term survival equals living donation.

	DBD	DCD
Circulation	Maintained	Withdrawn; recovery after death declared
Warm ischaemia	Minimal	Present → rapid aortic cannulation and cooling
Preservation	Static cold storage	Usually machine perfusion to reduce DGF

ALGORITHM

Procurement: midline incision + sternotomy → mobilise bowel off retroperitoneum → control aorta and inferior vena cava (IVC) → identify aberrant/iliac accessory arteries **before cannulation** → perfuse; ice via opened Gerota fascia → kidneys with perinephric fat, **periureteric tissue preserved** → en bloc removal → back-table separation, cold storage

- **Cold preservation at 4 °C** slows (not stops) metabolism; University of Wisconsin (UW) or histidine–tryptophan–ketoglutarate (HTK) solution. Hypothermic machine perfusion reduces DGF without improving long-term graft survival — it acts on ischaemia–reperfusion injury.

5 · LIVING DONATION

- Better graft and patient survival; planned, possibly pre-emptive operation; negligible transmission risk. **Donor safety comes first**; an independent donor advocate is mandatory.
- Exclusions: kidney disease, malignancy, active infection, cardiovascular disease, diabetes, significant obesity; **bilateral stones** generally exclude.
- Work-up: function by ≥2 methods (one measured), glycated haemoglobin, serology, cancer screening, **CT angiography** for vascular anatomy, split function if needed.

Counselling point	Figure
Long-term function	60–80% of baseline
Lifetime ESRD	90/10,000 vs 14/10,000 matched non-donors (general population 326)
Mortality	**3/10,000**
Pregnancy	More gestational hypertension/pre-eclampsia

- **Laparoscopic donor nephrectomy** (Ratner 1995) is standard; open reserved for complex anatomy. Principles: minimal handling, adequate vessel length, periureteric tissue preserved, short warm ischaemia.
- **Kidney paired donation**: ABO- or donor-specific antibody (DSA)-incompatible pairs exchange donors through registries.

6 · THE OPERATION

Preparation

- Consent: DGF, primary non-function, thrombosis, bleeding, urinary complications, donor-derived disease. ABO verification before induction; antibiotics; central line; Foley accessible for bladder filling.
- Back table: ligate vein tributaries and hilar lymphatics (lymphocele); handle artery gently (intima); **never dissect the triangle between lower pole and ureter**. Deceased right kidney: extend the short vein with donor IVC.
- **Multiple arteries** (18–43%): more DGF, similar survival. Options: long Carrel patch, side-to-side conjoining, end-to-side reimplantation, separate anastomoses. **Always keep the lower pole artery** (it supplies the ureter); upper pole branches ≤2 mm may be ligated.

Site and incision

- Retroperitoneal iliac fossa; right traditional (iliac vein easily mobilised). **Contralateral placement** brings pelvis and ureter anterior, easing later reconstruction.

	Gibson	Hockey stick
Course	Oblique, 2 cm above pubis to near anterior superior iliac spine (ASIS)	Midline suprapubic, up rectus edge
Access	Iliac vessels	Better to common iliac, IVC, peritoneum
Hernia	Lower	Higher; rectus denervation

- Pitfalls: incision near ASIS injures lateral femoral cutaneous nerve; retractor blade on femoral nerve; genitofemoral nerve. Divide inferior epigastrics; preserve the cord; ligate perivascular lymphatics.

ALGORITHM
Vein first (shortens arterial ischaemia): end-to-side to external iliac vein, valves excised, 5-0/6-0 polypropylene → artery end-to-side to external iliac (punch arteriotomy, avoid calcified clamp sites) → bulldogs, release iliac clamps, haemostasis → reperfuse → position without kink or stretch

- **Hume test**: finger compression of the renal vein → pulsatile swelling, relieved on release.
- Flaccid or tense engorged graft after reperfusion → check inflow, outflow, twisting; intraoperative ultrasound; redo if needed.
- Avoid the internal iliac artery if the contralateral one is occluded (gluteal claudication, erectile dysfunction).

Ureteric anastomosis

ALGORITHM
Lich–Gregoir extravesical: fill bladder → myotomy to mucosa → open mucosa distally → spatulate ureter → mucosa-to-mucosa 5-0/6-0 absorbable ± stent, distal anchoring stitch → close detrusor loosely as short tunnel

- Alternatives: ureteroureterostomy, pyeloureterostomy, to conduit. Document stent removal plan — forgotten stents encrust. Urological complications → *Ch 92*; paediatric recipients → *Ch 92/61*.

7 · SURGICAL COMPLICATIONS

- **Bleeding**: brisk drain output, falling haematocrit → transfuse and re-explore.

	Arterial thrombosis	Venous thrombosis	Renal artery stenosis
Picture	Sudden anuria, early	Swollen painful graft, haematuria, oliguria	Refractory hypertension, fluid overload, flash pulmonary oedema, falling function
Doppler	No signal	**Reversed diastolic flow**	High velocity; **parvus-tardus** distally
Treatment	Immediate exploration; nephrectomy if unsalvageable	Exploration, thrombectomy; nephrectomy if necrotic	**Angioplasty/stent** first; surgery risks graft

- Thrombosis risks: extremes of age, cold ischaemia >24 h, technical error, thrombophilia (factor V Leiden, protein C/S, antiphospholipid) → anticoagulate selected patients.
- **Torsion**: intraperitoneal grafts only → nephropexy. **Pseudoaneurysm** (<1%, often mycotic) → nephrectomy and repair. Biopsy arteriovenous fistula with steal → coil embolisation.

Lymphocele

- Risks: extensive dissection, retransplant, obesity, anticoagulation, mechanistic target of rapamycin (mTOR) inhibitors. May compress ureter, iliac vein, bladder.
- Aspirate to distinguish from urinoma †(lymph creatinine ≈ serum; urinoma creatinine far higher).
- Treat only if symptomatic: **laparoscopic peritoneal fenestration** most successful; sclerotherapy or drain intermediate; aspiration alone worst. Prevent by lymphatic ligation.

Transplant nephrectomy

- Indications: **graft intolerance syndrome** (pain, swelling, haematuria, fever — steroids first), infection, tumour. Not for asymptomatic failed grafts.
- **Early (<6–8 weeks) extracapsular**; late **subcapsular** (pedicle at hilum) — except tumour.

8 · REJECTION

- Typical regimen: induction, then tacrolimus + mycophenolic acid ± steroids. Acute rejection fell from ≥60% to <10%; long-term loss is mainly **chronic antibody-mediated injury**, subclinical until late.

Type	Mechanism
Hyperacute	Preformed DSA → immediate failure; prevented by crossmatch. †Irreversible → graft nephrectomy
Acute	Acute functional decline
Subclinical	Histology without dysfunction
Chronic	Progressive decline, chronic changes

- **Biopsy is required**; Banff classification: T-cell-mediated (TCMR) and antibody-mediated (AMR).
- Risks: HLA mismatch, DSA, young recipient/old donor, long ischaemia, DGF, non-adherence, retransplant.
- Monitoring: creatinine/eGFR; **de novo DSA** predicts graft loss (linked to class II mismatch and under-immunosuppression); donor-derived cell-free DNA (dd-cfDNA) has negative predictive value >90%, avoiding biopsy.

	TCMR	AMR
First line	Intravenous steroids	Steroids, intravenous immunoglobulin, plasmapheresis, rituximab; eculizumab selectively
Escalation	Antithymocyte globulin (Banff II–III)	Anti-interleukin-6 (chronic active)

9 · LANDMARK TRIALS

Trial	Design	Result	Take-home
Wolfe 1999	Registry: transplant vs wait-listed dialysis	Risk 2.84 early; equal at 106 days; long-term 0.32	Early peril, durable survival gain

10 · MCQ ONE-LINERS

1. ESRD: KDIGO G5, eGFR <15.
2. Deceased-donor listing at eGFR ≤20.
3. Genetic FSGS does not recur; primary does.
4. Primary hyperoxaluria: liver–kidney transplant.
5. Live vaccines contraindicated after transplant.
6. Incidental renal tumour <1 cm: no treatment needed first.
7. O universal donor; AB universal recipient.
8. Class II HLA: DR, DQ, DP.
9. Hyperacute rejection: preformed DSA.
0. En bloc for donors <15 kg.
1. Machine perfusion reduces DGF only.
2. Donor mortality 3/10,000.
13. First laparoscopic donor nephrectomy: Ratner, 1995.
14. Venous anastomosis before arterial.
15. Lower pole artery supplies ureter.
16. Commonest ureteric anastomosis: Lich–Gregoir.
17. Venous thrombosis Doppler: reversed diastolic flow.
18. Stenosis: parvus-tardus; angioplasty first.
19. Best lymphocele treatment: laparoscopic fenestration.
20. Late transplant nephrectomy: subcapsular.
21. TCMR first line: intravenous steroids.
22. dd-cfDNA negative predictive value >90%.
23. Gibson incision: lower hernia rate than hockey stick.
24. Torsion occurs only with intraperitoneal grafts.
25. Hume test: renal vein occlusion → pulsatile swelling.

92 / 61 Urologic Complications of Renal Transplantation and Paediatric Transplant Considerations

Reading time ≈ 33 min (slow pace)

1 · OVERVIEW

- Renal transplantation is the most cost-effective treatment for end-stage renal disease (ESRD). Donor/recipient selection, vascular technique and immunosuppression → *Ch 91*.
- Urologic complications are the **commonest surgical adverse event** (**8–9.5%**). Most arise from the **transplant ureter and ureterovesical anastomosis** — this organises the whole subject.
- **Allograft denervation** abolishes colic: obstruction, encrusted stents and stones present silently as falling urine output, rising creatinine or recurrent urinary tract infection (UTI).

2 · HAEMATURIA AND URETERAL STENTS

Haematuria

- Immediate postoperative: from the ureterovesical anastomosis; self-limiting over days, rarely needs irrigation; commoner after **Leadbetter** anastomosis.
- Persistent after graft biopsy = **arteriovenous–collecting system fistula** until proven otherwise: duplex ultrasound diagnoses, magnetic resonance angiography confirms. Most close spontaneously; others → highly selective embolisation.
- Asymptomatic microhaematuria: American Urological Association (AUA) work-up **plus BK polyomavirus** — suspected from cytopathic changes on cytology, confirmed by urinary BK titres.

Stent timing

- Randomised trials: routine stenting across the anastomosis **reduces leak and stricture**. Cost: bacterial colonisation in ~¼ of stents (independent of dwell time) → UTI, graft infection, graft failure.
- Meta-analyses: removal before week 3 reduces UTI without more leak or stenosis; optimum **14–21 days**. Removal **before 14 days increases leak** — hence a window, not a date.
- Individualise: surgeon's confidence in the anastomosis; diabetes mellitus, transplant revision, current UTI.
- Prophylaxis at removal: patients already on trimethoprim-sulfamethoxazole for *Pneumocystis jirovecii* pneumonia gain nothing from more; others may benefit from cover.

Retained / encrusted stent

- Mechanisms: stent caught by the ureteroneocystostomy suture (wait for dissolution or transect the stitch endoscopically), or **encrustation** (up to 5.7%, function of dwell time).
- Presents silently (denervation): declining function, falling output, recurrent UTI; rarely a vague ache over the graft.
- Prevention: a "Swiss cheese" chain of latent errors (undocumented insertion, poor communication, inadequate counselling). Stent cards, registries, apps — none fully reliable.

ALGORITHM

Computed tomography (CT) in every case (burden, distribution → approach) → **percutaneous nephrostomy first** (decompression, function recovery, antegrade access) → clearance by combined percutaneous nephrolithotomy (PCNL), retrograde intrarenal surgery (RIRS) and extracorporeal shock wave lithotripsy (ESWL)

3 · URINARY LEAK

- **Commonest early urologic complication** (1.2–8.9%).

- Dominant mechanism: **distal ureteric ischaemia** from compromised blood supply at procurement → necrosis. Risk: older donor, extended-criteria donor, multiple arteries, **failure to reimplant a lower polar artery** †(it often supplies the ureter).
- Other causes (Box 92.1) group into ischaemia, technical error and back-pressure: over-long ureter; premature removal of bladder/ureteric drainage; suture dehiscence, twisting, kinking; retention or blocked catheter; parenchymal necrosis; parenchymal perforation by a double-J stent. All preventable → surgeon-dependent rates.
- Diagnosis: with a drain — high output, falling urine output, rising creatinine; without — wound leakage/dehiscence, scrotal swelling, pain. Ultrasound or non-contrast CT finds the collection; **fluid creatinine far above serum** proves urine.

Management

- Principle: divert urine away from the extravasation — **Foley catheter + percutaneous nephrostomy**. The nephrostomy is diagnostic (antegrade nephrostogram) and therapeutic (antegrade stent). Retrograde stenting is harder (angulation, orifice oedema, risk of disrupting the anastomosis).
- Percutaneous drain for a large urinoma, obstruction by it, or infection.
- Small anastomotic leak: contrast studies every 1–2 weeks; remove catheter and nephrostomy once healed; follow for later stenosis.
- **Explore early** for a large proximal leak or one persisting despite maximal drainage: conservative success is low, adhesions have not formed, fibrosis is minimal, later stenosis less likely.
- Repair: tension-free ureteroneocystostomy if the remnant is well vascularised; otherwise ureteroureterostomy to the ipsilateral native ureter, bladder flap, or pyelovesicostomy.

4 · URETERAL OBSTRUCTION AND STRICTURE

Early (<3 months) — technical, reversible	Late (>3 months) — ischaemic, metabolic, drug
Technical error at ureteroneocystostomy	Stricture (predominant)
No stent; anastomotic oedema; redundant ureter	Stones
Extrinsic: lymphocele, haematoma, abscess	Lymphocele; immunosuppressant-related fibrosis

> **KEY POINT** Lymphocele spans both periods. Trap: a late obstruction is not automatically a stricture — exclude stones and lymphocele.

- Stricture: **0.5–9%**; primary cause **ischaemia**. Risks are perfusion or healing insults: cold ischaemia >435 min, acute rejection, urinoma/abscess, delayed graft function (DGF), multiple arteries, older donor, and transplant-specific **BK virus and cytomegalovirus (CMV)**.
- No difference by anastomotic technique, or brain-dead vs donation-after-cardiac-death donors.
- Suspect with declining function, especially with hydronephrosis. Dilatation may be **less than expected** because periureteric fibrosis stops expansion → antegrade nephrostogram or Whitaker test.

Treatment

- First decompress (nephrostomy, nephroureteral or double-J stent). **Antegrade preferred**: retrograde access is hampered by neo-orifice position and tortuosity; nephrostomy gives detailed imaging and instrument access. Once stented, retrograde exchange is easy — no long-term nephrostomy needed.
- Endoscopic ranking: stent alone poor < balloon dilation (most studied, moderately durable) < endoureterotomy (holmium laser > Acucise > resectoscope T-loop).
- **Open repair is most effective long term**: reimplantation, ureteroureterostomy to native ureter, Boari flap, appendiceal/ileal interposition, pyelovesicostomy. Robotic repair comparable short term; indocyanine green shows ureteric viability.

5 · VESICOURETERAL REFLUX IN ADULTS

- Reflux on voiding cystography in **50–86%**, but clinically significant in **<1%**, independent of technique — antireflux modifications change the radiology, not the clinical problem.
- Causes: technical error, bladder outlet obstruction, high-pressure storage (detrusor overactivity, poor compliance).
- Recurrent UTI → voiding cystography → exclude outlet obstruction and high-pressure storage. Treat the **cause** first.
- Endoscopic injection (Teflon, Macroplastique) only for significant grade I–II. High grade or failed injection → revision or reimplantation into ipsilateral native ureter.

6 · LYMPHOCELE

- Lymph pseudocyst in a fibrous capsule; most within 6 months, **peak 6 weeks**. Cause: dissection of lymphatics around recipient iliac vessels or donor renal vessels (procurement/back table).
- Associations: obesity, age, dialysis duration, warm ischaemia, prophylactic low-molecular-weight heparin, DGF, rejection, re-transplantation, **mammalian target of rapamycin (mTOR) inhibitors**.
- Large ones compress: limb oedema, graft dysfunction, bladder compression, fever, deep vein thrombosis (external iliac vein).
- Fluid biochemistry and microbiology separate it from urinoma, seroma, abscess.
- **<15%** need treatment. Efficacy: aspiration alone worst < drain ± sclerotherapy < open surgery < **laparoscopic peritoneal window** (best). †The window drains lymph into the peritoneum, where it is absorbed.
- Open preferred with wound complications, or a small lymphocele beside vital renal structures.

7 · STONES AND UTI

- Stones ~1%, mean onset ~28 months; calcium > struvite > uric acid. Factors: secondary hyperparathyroidism, UTI, hypercalciuria, hyperuricosuria, hypocitraturia, hyperoxaluria. **Ciclosporin → hyperuricaemia; calcineurin inhibitors → hyperoxaluria, hypocitraturia.**
- Native-kidney algorithm: non-obstructing asymptomatic <4 mm → ultrasound and creatinine surveillance; growing/symptomatic → ESWL, RIRS, PCNL. Afterwards watch for silent hydronephrosis; metabolic screen.
- UTI: commonest infection, 30% of sepsis admissions, highest in first 6 months; more in women by 3 years. Risks: prolonged stent, reflux, rejection, poor function, high prednisolone dose.
- Screening/treating asymptomatic bacteriuria beyond 2 months: unsupported, except with **unexplained creatinine rise**.
- Cystitis: fluoroquinolone, amoxicillin-clavulanate, third-generation cephalosporin or nitrofurantoin (creatinine clearance >30 mL/min) for **5–10 days**. Pyelonephritis/complicated: **14–21 days**, and confirm the graft drains.
- Recurrent UTI (≥3/12 months) → structural evaluation (reflux, bladder dysfunction, outflow obstruction); vaginal oestrogen if postmenopausal.

8 · LOWER URINARY TRACT, SEXUAL FUNCTION, MALIGNANCY

- Voiding dysfunction is **concealed by oliguria** and declares itself once output returns. Contributors: capacity <300 mL, anatomic abnormality, immunosuppression reducing bladder mass. Evaluate with history, examination, diary; urodynamics/cystoscopy selectively.
- **Small defunctionalised adult bladder regains normal capacity** — pretransplant cycling or augmentation unnecessary. Symptomatic: anticholinergic → botulinum toxin → augmentation/conduit.
- Benign prostatic hyperplasia (BPH): 3-year incidence 9.7%, linked to UTI and graft loss. α-blocker, treat constipation before catheter removal, selective suprapubic tube.

- **Avoid prostate resection while anuric** — healing without urine flow → bladder neck contracture, stricture. Transurethral resection (TURP) or holmium enucleation is safe **≥3 weeks** post-transplant; never in the first 2 weeks or with a stent (sepsis, deaths).
- Stress incontinence managed as usual; artificial sphincter feasible.
- Erectile dysfunction: 87.5% in ESRD; 20–50% persist after transplant; earlier transplantation does better (pre-empts vasculopathy). Standard treatment. Testosterone improves after transplant.

Cancer	Key points
Renal cell carcinoma (RCC)	Above general risk; **90% in native kidneys**; graft tumours mostly papillary/low grade. Metastatic graft-origin: some advise nephrectomy and stopping immunosuppression to reject donor cells. Kidney Disease: Improving Global Outcomes (KDIGO) does not endorse routine native screening — only high risk (previous RCC, analgesic nephropathy, tuberous sclerosis, acquired cystic disease)
Prostate	Incidence as general population; graft proximity and healing complicate therapy; surgery > radiotherapy > surveillance (underused); equal oncologic outcomes
Bladder	**~3-fold risk**, more aggressive → intensive surveillance; bacillus Calmette-Guérin (BCG) usable in selected high-risk non-muscle-invasive disease; muscle-invasive: chemoradiotherapy or cystectomy protecting graft vessels and ureter

9 · PAEDIATRIC: PRINCIPLES AND PRETRANSPLANT ASSESSMENT

- Children needing kidney replacement therapy (KRT) often have **congenital anomalies of the kidney and urinary tract (CAKUT)**. Commonest age 13–17 years; congenital: hypo/a/dysplasia, obstructive uropathy (chiefly posterior urethral valves (PUV) in young boys); acquired: focal segmental glomerulosclerosis.
- Task: the condition that damaged the native kidneys must not damage the graft. Involve urology **before end-stage kidney disease (ESKD)**; referral after the graft is in jeopardy is suboptimal.
- **Five goals**: unimpeded drainage; capacious low-pressure reservoir with social continence; volitional emptying; no infection; fewest procedures.
- Screen all: history, ultrasound, uroflow + postvoid residual. Voiding cystourethrography (VCUG) not routine — for neuropathic bladder, obstruction, vesicoureteral reflux (VUR), prior surgery, febrile UTIs, unexplained hydroureteronephrosis.
- Urodynamics: neuropathic bladder, PUV, large ureterocele, refractory dysfunction, high-grade hydronephrosis, recurrent UTI; repeat over time.

10 · PAEDIATRIC PRETRANSPLANT PREPARATION

- Typical ESKD bladder: low capacity, overactivity, poor compliance — dangerous because **high pressure reaches the kidney even without reflux**. PUV bladders evolve towards **myogenic failure**.
- Target: age-expected capacity at detrusor pressure **<u><40 cm H_2O></u>**. First line antimuscarinics (oxybutynin; then solifenacin, mirabegron, intradetrusor botulinum toxin) with clean intermittent catheterisation (CIC). Prove emptying by repeated low residuals.
- Emptying matters as much as storage; treating compliance can impair emptying and force CIC. Sensate or difficult urethra → **continent catheterisable stoma**.
- **CIC adherence before transplant tests adherence** to post-transplant demands.
- Infections persist after transplant — resolve stasis beforehand (double voiding, CIC). Separate UTI from colonisation; persistent infection → diverticula or ureteral stumps.

Defunctionalised bladder

ALGORITHM
Catheterise → instil increasing saline volumes → set dwell → drain → step up volumes (± medication) → target capacity for age → **confirm by urodynamics** → clear for transplant; inadequate → augment

KEY POINT A renal allograft must never be used to test whether a bladder works.

Augmentation

- Indications: capacity **<75%** of expected, pressures **>40 cm H_2O**, CIC needed every **≤3–4 h** despite maximal therapy. Augmentation does not clearly raise transplant risk.
- Timing: PUV — postpone (myogenic failure); borderline neuropathic bladder — earlier, as robust graft output may decompensate it.
- Ileum/sigmoid commonest; lifelong malignancy surveillance. **Gastric** now rare: haematuria–dysuria syndrome, worst in anuric children. **Ureterocystoplasty** preferred if a dilated ureter exists (no mucus, stones, metabolic load) — so preserve ureters.
- Stomas medial to the future Gibson incision; children <15 kg may need aorto-caval anastomosis transperitoneally. A functioning ileal loop can become the augment.

11 · PAEDIATRIC RECONSTRUCTION AND NATIVE NEPHRECTOMY

- Peritoneal dialysis (PD): intraperitoneal surgery usually forces temporary haemodialysis.
- Major reconstruction **before transplant**; allow **6–12 weeks** after enterocystoplasty. Exception: transplant in first 2 years, before continence matters. Document with a diagram.
- Avoid concurrent procedures in infants (anaesthesia time, instability, anticoagulation bleeding); single ipsilateral nephrectomy through the transplant incision for space is acceptable.
- **Native nephrectomy** (default: leave): refractory hypertension, nephrotic syndrome with malnutrition/hypercoagulability, massive polyuria, recurrent upper tract infection, large stones, malignancy risk, very large kidneys, massive VUR. Retained kidneys provide water excretion, erythropoietin and vitamin D.
- On PD: retroperitoneal approach (prone for bilateral); not on PD: any approach; embolisation for high risk.
- Ureters: transect high; leave a normal ureter (salvage ureteroureterostomy); remove a dilated refluxing one. Discuss circumcision for boys with valves.

12 · PAEDIATRIC TRANSPLANT URETER

- Preferred: **extravesical non-refluxing ureteroneocystostomy** †(Lich–Gregoir type).

ALGORITHM
Partially fill bladder → traction sutures → size ureter → detrusor incision **3–4 cm** → excise distal mucosal disc → drain bladder → spatulate ureter 4–5 mm → interrupted mucosa-to-mucosa fine absorbable → anchor adventitia (prevents eversion) → close detrusor loosely at entry

- Barry: 4 cm tunnel between parallel incisions. Ureteroureterostomy to native ureter is an alternative and the standard **salvage** for stenosis.
- Stenting not routinely supported; use for difficult reimplant, abnormal bladder, damaged ureter. Short double-J out at 4–6 weeks, or early (5–7 days) via urethral catheter tether.

13 · PAEDIATRIC COMPLICATIONS

- Ureteral stenosis 6%, leak 3%, stones 2%, significant reflux 2%; higher with pre-existing uropathy. **Thrombosis** is a leading cause of early graft loss.
- **Leak**: check/irrigate catheter → ultrasound → mercaptoacetyltriglycine (MAG3) renogram or cystogram. Use the **largest catheter the urethra accepts**. Large/persistent →

nephrostomy/exploration; necrotic ureter replaced by reimplant, native ureter, psoas hitch or flap.

- **Reflux into graft** is distinct (reimplanted ureter, immunosuppression). Recurrent febrile UTI → reimplantation; asymptomatic → observe; endoscopic role limited.
- **Hydronephrosis**: >50% of obstructions within 100 days. Catheter first; improvement points to bladder dysfunction.

ALGORITHM

Rising creatinine + hydronephrosis → rejection signs → biopsy, treat rejection → no rejection and unimproved by bladder drainage → stent/nephrostomy → failed retrograde → percutaneous antegrade study and stent → focal stenosis: balloon + stent 4–6 weeks → open repair without delay

- Bladder dysfunction: 1–2 weeks catheter drainage; falling creatinine confirms. Stones: intervene >1 cm, growing, or obstructing/infected; choose single-session modality.

14 · MCQ ONE-LINERS

1. Commonest early urologic complication: urinary leak.
2. Predominant late (>3 months) obstruction cause: ureteral stricture.
3. Primary cause of transplant stricture: ischaemia.
4. Optimal stent removal: 14–21 days; earlier increases leak.
5. Persistent haematuria after biopsy: arteriovenous fistula.
6. Urine leak proof: fluid creatinine far above serum.
7. Lymphocele peak: 6 weeks.
8. Best lymphocele treatment: laparoscopic peritoneal window.
9. mTOR inhibitors associated with lymphocele.
10. Reflux into adult graft: 50–86% radiologically, <1% significant.
11. TURP safe ≥3 weeks post-transplant, not with stent.
12. Post-transplant RCC: 90% in native kidneys.
13. Bladder cancer risk ~3-fold, more aggressive.
14. Paediatric target pressure: <40 cm H_2O.
15. Augment if capacity <75% expected for age.
16. Gastric augmentation: haematuria–dysuria syndrome.
17. Allow 6–12 weeks after enterocystoplasty before transplant.
18. Preferred paediatric anastomosis: extravesical non-refluxing ureteroneocystostomy.
19. Commonest paediatric urologic complication: ureteral stenosis (6%).
20. Use largest urethral catheter at transplantation.
21. >50% paediatric obstructions occur within 100 days.

★ Putting it together: kidney function and transplant decisions

Reading time ≈ 7 min (slow pace) · Integrates Ch 88, 90–92 (with 61) · use these as the skeleton of long-essay answers

A · RISING CREATININE AFTER UROLOGICAL SURGERY

ALGORITHM

1. Postrenal first: check the catheter, scan the bladder and kidneys (hydronephrosis?). After cystectomy or bladder repair, rising creatinine with ascites may be an intraperitoneal urine leak reabsorbed across the peritoneum — drain fluid creatinine far above serum confirms it (→ *Ch 90*).

2. Prerenal or intrinsic: fractional excretion of sodium (FeNa) <1% and bland urine → volume depletion, recovers with fluids; >1% with granular casts → acute tubular necrosis. Review nephrotoxins and contrast; stop renin–angiotensin blockers and metformin while unwell.

3. Dialysis only for "AEIOU" that fails medical treatment: refractory acidosis, electrolytes (potassium), intoxication, overload, uraemic complications.

B · POST-OBSTRUCTIVE DIURESIS

KEY POINT After relieving bilateral obstruction (or obstruction of a solitary kidney), monitor urine output, electrolytes and weight. Most of the diuresis is physiological, clearing retained salt and water; † it becomes pathological when it continues after the patient is back to normal volume. † Do not replace every millilitre — that keeps the diuresis going. If the patient cannot drink, replace about 75% of output with 0.45% saline and adjust to electrolytes (→ *Ch 90*).

C · TRANSPLANT URETERIC COMPLICATIONS

Problem	Clue	Management
Urine leak (early)	High drain output, falling urine output; fluid creatinine far above serum	Foley catheter + percutaneous nephrostomy (± antegrade stent); explore early if large, proximal or persistent
Ureteric stricture	Declining function ± hydronephrosis (may be less dilated than expected)	Nephrostomy and antegrade study → balloon dilation or endoureterotomy; open repair (reimplant, ureteroureterostomy to native ureter, Boari flap) is most durable
Lymphocele	Fluid creatinine ≈ serum; may obstruct ureter or compress vessels	Treat only if symptomatic (<15%); laparoscopic peritoneal window is best; aspiration alone recurs
Stent-related urinary tract infection	Colonised stent	Remove the routine stent at 14–21 days (earlier → leak; later → infection)

Prevention: routine stenting across the anastomosis, never dissecting the triangle between the lower pole and the ureter on the back table, and reimplanting lower polar arteries (→ *Ch 91, 92/61*).

D · REJECTION AT A GLANCE

Type	Cause	Point
Hyperacute	Preformed donor-specific antibody	Immediate failure; prevented by crossmatch; † irreversible → graft nephrectomy
Acute	T-cell-mediated (TCMR) or antibody-mediated (AMR), Banff classification	Biopsy required. TCMR: intravenous steroids, then antithymocyte globulin. AMR: steroids, intravenous immunoglobulin, plasmapheresis, rituximab

Type	Cause	Point
Chronic	Mainly antibody-mediated injury	Main cause of late graft loss; subclinical until late

E · CHILD GOING FOR TRANSPLANT

ALGORITHM

Most have congenital anomalies of the kidney and urinary tract, chiefly posterior urethral valves. Before listing, make the bladder safe: capacity appropriate for age at detrusor pressure <40 cm H_2O, using antimuscarinics and clean intermittent catheterisation, and augmentation if needed. Remove dilated refluxing native ureters or kidneys causing infection, polyuria or hypertension. A poor bladder left untreated destroys the graft (→ *Ch 92/61*).